SAFARI LANKA

Web Based Wildlife Safari Trip Management System

Final Changes, Implemented Features, Full Workflow and Recommendations

Consolidated final project documentation - 15 October 2026

Current stack: React 18 + Vite frontend, Java 17 + Spring Boot 3 backend, Spring Security/JWT, JPA/Hibernate and MySQL 8+.

This document consolidates the current codebase with the major Safari Lanka requirements and refinements requested across the project work. It separates implemented features from recommendations so the final academic prototype can be explained clearly during demonstrations and evaluations.

1. Executive Summary

Safari Lanka is a role-based wildlife safari management system for Yala National Park. The current build covers the complete tourist journey from package discovery and booking through payment, resource assignment, permit issue, park entry, safari execution, park exit, trip completion and post-trip review. Supporting modules cover finance, fleet/maintenance, guide/driver rosters and leave, park compliance, incidents, contact requests, audit logs, multilingual preferences, currency display, notifications, dark/light theme, an Admin GPS map and the Kavi 3D assistant.

Latest update applied in this final build

- Kavi now faces the visitor directly. Its body and head do not rotate with the mouse. The raised hand continuously performs a friendly hello wave and a small "Hi!" bubble is shown while idle.
- Dark-mode visibility on the Sign In page was strengthened, especially the Demo logins role names, email addresses, helper text and controls. Additional muted-text contrast hardening was applied to other public/dashboard areas.
- Children below 5 years are free of the package per-person charge. Adults and children aged 5-11 are chargeable. Under-5 children still count toward the six-person vehicle capacity and can therefore still affect the number of vehicles required.
- The Park Officer tab previously labelled "Gate Compliance CRUD" is now displayed as "Park Compliance". Existing API/entity/table names are preserved for compatibility.
- Previously added centered action-result popups and clickable role notifications remain enabled.

Important pricing rule

Booking amount = (Adults + Children age 5-11) x package price per person + additional vehicle surcharge. Children below 5 do not add a per-person charge, but they do count when deciding how many vehicles are needed.

2. System Architecture and Roles

The project uses a separated React frontend and Spring Boot REST backend. Authentication is JWT-based, business rules are enforced in the backend, and MySQL stores operational records. The Windows launcher starts both backend and frontend so the user can run the complete system with one project command.

Layer	Technology / responsibility
Frontend	React 18, Vite, React Router, Axios, Lucide icons, QR rendering, responsive public pages and role dashboards.
Backend	Java 17, Spring Boot 3.4.x, Spring Web, Spring Security, JWT, validation, JPA/Hibernate and business-rule enforcement.
Database	MySQL 8+ with users, bookings, packages, payments, permits, assignments, fleet, rosters, leave, compliance, incidents, reviews, messages, notifications, settings and audit logs.
Launcher	Root npm scripts + Windows BAT files for one-click development startup, database setup and optional ngrok/LAN demo access.

Role	Main responsibilities
Tourist / Visitor	Register, manage profile, browse packages, create/edit eligible bookings, pay after confirmation, view trip/permit/team information, cancel where allowed and submit reviews after completion.
Safari Operations Manager	Review/confirm/cancel bookings, manage packages, assign guide/driver/vehicle after payment, avoid conflicts, print booking summaries, review incidents and operational data.
Safari Guide / Naturalist	View assigned safaris, manage availability/leave, see special requirements, participate in trip operations and submit incident reports.
Jeep Driver	View assigned trips and vehicle, maintenance information, leave requests, perform trip work, and manually complete a trip only after Park Officer checkout.

Role	Main responsibilities
Authorized Park / Gate Officer	Review/issue/reject permits, validate check-in/check-out, manage Park Compliance records and view safari compliance incidents.
Finance & Administration Manager	Review unpaid/pending payments, approve/reject bank-transfer slips with reasons, manage finance/refund information and reports.
System Administrator	Manage users/logins/roles, vehicles and system settings; review bookings, incidents, contact requests, leave, audit activity and Admin GPS monitoring.

3. Consolidated Implemented Change Register

Public website and branding

- Premium Safari Lanka/Yala visual styling with local logo, transparent/glass public navigation, hero/background media and responsive layouts.
- Public navigation organized around Experience, About Yala and Contact Us, with scroll behavior refinements and theme support.
- About Yala block-based explorer with supporting images/details and premium content layout.
- Experience/review area for completed safari feedback, including driver/guide ratings and optional media.
- Contact Us page with local visual treatment; submitted contact requests are available to Administration.

Authentication, account and preferences

- Seven role-based demo logins plus tourist registration.
- My Account is available from dashboards with profile image support and editable profile data.
- Country, gender, birthday/age, preferred language and display currency support; language can be changed later.
- Dark/light theme toggle across public pages and dashboards, with repeated contrast improvements for dark mode.
- Admin can create role accounts using role-specific fields and role email policies; guide account creation does not request a driving licence, while drivers do.

Booking and visitor handling

- No same-day bookings; date must be tomorrow or later.
- Booking form simplified to Adults, Children 5-11, Children below 5 and Safari Date; new bookings no longer require individual visitor PII.
- Maximum 6 visitors per vehicle. Groups above 6 require additional vehicles within package capacity.
- Children below 5 are now free of the package per-person charge, while still counting for vehicle capacity.
- Special requirements are stored and visible to relevant operational roles.
- Tourists can edit only pending/unpaid eligible bookings and the backend recalculates capacity, vehicles and price.

Booking confirmation and payment

- New bookings remain PENDING until Safari Operations confirms them. Tourist payment is locked before confirmation.
- Card workflow validates an exact 16-digit card number with four-digit grouping, expiry month/year rules and secure demo handling.
- Bank transfer requires payment-slip upload and Finance approval. Operator assignment stays blocked until Finance approval succeeds.
- Finance rejection reason is stored and displayed in relevant tourist/operator views.
- Booking cancellation reason is stored and displayed when Operations cancels a booking.
- Finance dashboard includes unpaid/pending payment visibility and payment notifications.

Assignment, fleet and maintenance

- Assignment is permitted only after successful/approved payment.
- Operator assigns driver/guide; the driver-linked vehicle is automatically included in the trip assignment workflow.
- Assigned driver, guide and vehicle are prevented from conflicting assignment on the same date.

- Admin can add vehicles, link vehicles to drivers and change driver vehicle assignments.
- Driver and Operations dashboards include vehicle maintenance reporting/history and maintenance completion workflows.

Leave, roster and availability

- Guides and drivers can request leave; Admin can approve/reject it.
- Approved leave blocks conflicting trip assignment.
- Guide roster CRUD/availability records support planning and demonstration of member functionality.

Permit, park entry and compliance

- Permit record is linked to booking and reviewed only after confirmed booking, paid payment and required assignment readiness.
- Park/Gate Officer can approve and issue or reject permits with a recorded reason and reviewer details.
- Clicking a permit ID can load/copy it into digital permit validation.
- Check-in records park entry and moves the trip to an ongoing state. Check-out records park exit.
- Park Compliance provides draft create/read/update/delete inspection records and finalized read-only compliance history.
- UI label is now Park Compliance instead of Gate Compliance CRUD; backend data identifiers remain compatible.

Trip status and completion

- Trip Status becomes meaningful only after payment, assignment and permit readiness.
- Tourist can see assignment and gate progress; ongoing state follows Park Officer check-in.
- Driver cannot complete a safari before Park Officer checkout. Before checkout the driver sees a waiting message and the backend prevents bypass.
- After checkout, the driver manually marks the trip Completed; status then propagates to relevant dashboards.

Incidents, reviews, support and notifications

- Guide/driver incident reporting is visible to Operations and Administration.
- Completed tourists can publish verified reviews and ratings.
- Contact requests are stored and visible to Admin.
- Notifications exist for all roles with unread counts; notification cards are clickable and open full-message popups with related destination actions.
- Global success/error activity feedback uses centered popups after important actions such as check-in/out, approvals, rejections, saves and other workflow actions.

Admin tools and live map

- Administration includes login management by role, user detail/settings, booking management, incident reports, contact requests, system settings and audit activity.
- Admin GPS mini-map/full map supports Yala blocks/gates and moving safari vehicles with driver, guide, vehicle, package, booking and permit context.
- Map supports theme-aware visibility, zoom behavior and slower vehicle motion appropriate for demonstration.

Kavi 3D Safari Assistant

- Floating 3D-style Safari Lanka bot integrated into the real React site.
- Click opens a Safari-specific assistant panel with package, booking, payment, bank-transfer, trip status, permit, capacity and support guidance.
- Current final behavior: bot faces directly forward with no body/head cursor rotation; raised hand waves hello; small Hi bubble appears while idle.
- Assistant keeps quick actions to real site routes and explains key project workflow rules.

Validation and quality-of-life improvements

- Sri Lankan NIC support including legacy V format where applicable.
- Driver licence normalization/validation supports project-approved old/new Sri Lankan formats and blocks invalid saves on frontend/backend.
- Phone, email, password, country/NIC-passport, age and related registration validations are included.
- Search by booking ID and dashboard search fields are provided in operational views.
- Operator can print booking summary; package create/edit supports local images/videos and package deletion.

4. Full End-to-End Website Workflow

The following is the recommended sequence to explain during a system demonstration or viva. Each stage shows the actor, action and backend gate that must be satisfied.

#	Stage / actor	What happens	Gate / result
1	Public visitor	Open home, browse packages, About Yala, Experience and Contact.	No login required for public discovery.
2	Tourist registration / sign in	Create account or use demo login. Profile stores country/language/currency and account information.	Authenticated Tourist role.
3	Tourist booking	Choose package, future date, Adults, Child 5-11 and Child below 5; add contacts/special requirements.	Same-day blocked. Capacity checked. Under-5 is free per-person. Booking = PENDING, payment = UNPAID.
4	Safari Operations	Review booking and confirm or cancel with a reason.	If confirmed, booking = CONFIRMED and tourist payment unlocks.
5A	Tourist card payment	Enter valid demo card details.	On success payment becomes PAID.
5B	Tourist bank transfer	Upload slip.	Payment = PENDING until Finance review.
6	Finance Manager	Approve or reject bank transfer; rejection stores a reason.	Approved bank payment becomes PAID. Assignment is still blocked if not paid.
7	Safari Operations assignment	Assign available driver/vehicle and guide; avoid same-date conflicts and approved leave.	Required number of teams/vehicles must match booking vehicle count.
8	Park / Gate Officer permit review	Review paid confirmed booking with complete assignment and approve/issue or reject permit with reason.	Permit becomes APPROVED/issued.
9	Tourist trip status	View team, permit and trip progress.	Status data shown from payment + assignment + permit workflow.
10	Park Officer check-in	Validate digital permit at entry.	Check-in time recorded; safari/trip becomes Ongoing.
11	Guide + Driver trip execution	Conduct safari; relevant staff can see booking requirements. Guide/driver may report incidents.	Trip remains ongoing while tourists are inside park.
12	Park Officer check-out	Validate permit on exit.	Checkout time recorded; driver completion gate is unlocked.
13	Driver completion	Driver manually completes assigned safari.	Backend rejects completion before checkout; after checkout booking/trip can become Completed.
14	Tourist post-trip review	Tourist can add verified review/rating after successful completion.	Review is linked to completed trip.
15	Admin / Operations / Finance reporting	Review audits, incidents, contacts, compliance, payments, bookings, GPS and system records.	Historical operational trail remains available.

5. Detailed Role Workflows

Tourist / Visitor

- Browse packages -> create booking -> wait for Operations confirmation.
- After confirmation choose card or bank transfer.
- If bank transfer, wait for Finance approval; if rejected, read rejection reason and follow the allowed correction flow.
- After payment and assignment, view driver/guide/vehicle details.
- After permit issue, use trip/permit status; after check-in see ongoing progression.
- After trip completion, submit review/rating.

Safari Operations Manager

- Review pending bookings -> confirm or cancel with reason.
- Wait for PAID payment status before assignment.
- Assign available driver/vehicle and guide while respecting leave/date conflicts and required vehicle count.
- Manage safari packages and media; print booking summary.
- View incidents, special requirements and operational searches.

Finance & Administration Manager

- Monitor unpaid/pending bookings/payments.
- Review bank transfer slip -> approve or reject with reason.
- Approved payment sets booking payment status to PAID and unlocks Operations assignment.
- Handle finance/refund reporting where enabled.

Park / Gate Officer

- Review permit after confirmed + paid + assigned readiness.
- Approve/issue or reject with recorded decision details.
- Validate permit check-in -> park entry / ongoing.
- Validate permit check-out -> park exit and driver completion unlock.
- Manage Park Compliance draft inspections; finalize for read-only history.

Jeep Driver

- View assigned trip, guide, visitors, vehicle and requirements.
- See maintenance information and submit/track maintenance records as available.
- Cannot manually complete trip until Park Officer checkout.
- After checkout, complete trip manually; submit relevant incidents if needed.
- Use leave request workflow for unavailable dates.

Safari Guide / Naturalist

- View assigned trips and requirements.
- Manage roster/availability and leave requests.
- Participate in trip operations and submit incident reports after/around trip as designed.

System Administrator

- Manage users/logins/roles and role-specific account data.
- Manage vehicles/driver links, leave decisions and Administration tools.
- View booking/incident/contact/audit/system settings records.
- Use Admin Yala GPS map to monitor demo/live active safari context.

6. Critical Business Rules and Status Gates

Rule	Enforced behavior
Booking date	Visit date must be after today. Same-day booking is rejected.
Visitor capacity	All visitors, including under-5 children, count toward group size and vehicle capacity. Maximum 6 visitors per vehicle.
Under-5 pricing	Child below 5 = no package per-person charge. They can still increase required vehicle count and therefore vehicle surcharge.
Operator confirmation	Tourist cannot pay while booking is still PENDING.
Bank transfer	Finance approval is required before payment becomes PAID and before assignment.
Assignment	Only paid bookings can be assigned; same-date/resource conflicts and approved leave are respected.
Permit issue	Permit review/issue requires confirmed booking, successful payment and required assignment readiness.
Check-in	Park Officer check-in records park entry and starts ongoing trip state.
Driver completion	Driver completion is disabled and backend-blocked until Park Officer checkout exists.
Checkout	Park Officer checkout records exit and unlocks driver completion.
Cancellation / rejection reasons	Operator cancellation and Finance rejection reasons are persisted and shown to relevant users.
Review eligibility	Post-trip review is for successfully completed safaris.

7. Booking Price Examples

The examples below assume a package price of LKR 10,000 per chargeable person and an additional vehicle fee of LKR 8,000. They show the new under-5 rule clearly.

Group	Chargeable visitors	Vehicles	Calculation	Total
2 adults + 1 under 5	2	1	2 x 10,000	LKR 20,000
2 adults + 2 children 5-11 + 1 under 5	4	1	4 x 10,000	LKR 40,000
4 adults + 2 children 5-11 + 1 under 5	6	2	6 x 10,000 + 8,000	LKR 68,000
1 adult + 5 under 5	1	1	1 x 10,000	LKR 10,000

Capacity note: the under-5 visitor is free per person but still occupies a seat for safety and vehicle planning.

8. UI, Notifications and Kavi Assistant

Dark mode visibility

- Demo login buttons now use a dark surface with brighter role names and readable email addresses.
- Authentication helper text, separator text, eye icon and headings have stronger contrast.
- Additional muted-text rules improve booking summaries, price details, visitor controls, permit/review cards, list rows and dashboard supporting text.

Action feedback and notification behavior

- Important user actions can show centered success/error/information popups instead of relying only on a thin banner.

- Each role has notifications and unread counts. Clicking a notification opens a full-message popup.
- Where a destination is known, the popup can direct the user to the related dashboard tab/page.

Kavi final behavior

- Bot remains fixed in the floating assistant position.
- Head and body face forward; there is no cursor-follow rotation in the final version.
- Raised hand loops a friendly hello wave and a small Hi bubble appears when idle.
- Clicking Kavi opens Safari Lanka help with real quick actions and workflow-specific answers.

9. Database and Full SQL

The final SQL file is **database/SAFARI_LANKA_FULL.sql**. The final deliverable also includes a standalone copy named **Safari_Lanka_FINAL_FULL.sql** for easier import.

Main tables

Module	Tables
Identity / users	users, leave_requests, notifications
Packages / booking	safari_packages, bookings, visitors (legacy compatibility)
Fleet	vehicles, maintenance_records
Finance	payments, refund_requests
Permit / trip / park	permits, trip_assignments, guide_rosters, gate_compliance_records
Operational support	incidents, reviews, contact_messages
Administration	system_settings, audit_logs

Under-5 pricing requires no database schema migration because the age-group booking columns already exist. The full SQL now documents the rule and the booking overview view exposes a chargeable_visitor_count derived from adults + children 5-11.

Compatibility note: the visible UI name is Park Compliance, while the existing table/API naming such as gate_compliance_records is intentionally preserved to avoid breaking existing data and backend code.

10. Running the Final Project

Step	Action
1	Install Java JDK 17+, Node.js LTS/npm and MySQL 8+.
2	Import database/Safari_Lanka_FINAL_FULL.sql (or database/SAFARI_LANKA_FULL.sql) into MySQL. The project default local database configuration expects the project MySQL settings documented in README.
3	On Windows, use run.bat for the one-click development launcher. It starts backend and frontend and opens the Vite site.
4	Alternatively use the root npm scripts for dev backend/frontend workflow.
5	Use the seven demo accounts to demonstrate role-specific workflows. Do not use real payment credentials because the payment flow is academic/simulated.

Demo accounts

Role	Email	Password
Tourist	tourist@safarilanka.lk	Tourist@123
Safari Operator	operator@safarilanka.lk	Operator@123
Safari Guide	guide@safarilanka.lk	Guide@123
Jeep Driver	driver@safarilanka.lk	Driver@123
Park Officer	park@safarilanka.lk	Park@123
Finance Manager	finance@safarilanka.lk	Finance@123
System Administrator	admin@safarilanka.lk	Admin@123

11. Recommended Next Improvements

The current build is suitable as a feature-rich academic prototype. The following items are recommendations for stronger production-readiness or final-evaluation confidence; they are not required to use the delivered ZIP.

Priority	Suggestion	Why it helps
High	Add automated end-to-end tests for booking -> confirmation -> payment -> assignment -> permit -> check-in -> checkout -> driver completion.	Protects the most important cross-role workflow from regression.
High	Introduce Flyway or Liquibase migrations for future DB changes.	Makes upgrades repeatable and safer than manual SQL sequencing.
High	Add integration tests specifically for under-5 pricing and multi-vehicle edge cases.	Prevents mismatch between frontend estimate and backend total.
High	Add accessibility checks (keyboard, focus, contrast, labels) and mobile-width testing.	Improves usability and helps catch dark-mode/readability issues early.
Medium	For production only, replace demo payments with a tokenized payment gateway and never store card credentials.	Improves security/compliance beyond the academic simulation.
Medium	Move package/review media from local folders to managed object storage if deployment becomes multi-server.	Improves reliability and scalability.
Medium	Add email/SMS/push delivery as an optional layer on top of in-app notifications.	Users can receive important approval/rejection/trip updates when not logged in.
Medium	Add audit/export reports for payment decisions, permit decisions, assignments and compliance history.	Useful for evaluation, traceability and operations reporting.
Low	Connect Admin GPS to real telematics only if actual vehicle tracking is required.	The current moving-map behavior is appropriate for a campus demo; real GPS would be a future integration.

12. Verification Performed for This Final Update

- All 43 frontend JS/JSX source files were parsed successfully with Babel JSX/module parsing after the edits (0 syntax errors).
- The final Kavi composition was checked using the actual separated bot body/head/hand assets; the bot faces forward with the raised waving hand.
- The booking frontend and backend now use the same chargeable visitor rule: adults + children 5-11.
- The full SQL was updated to document the pricing policy and expose chargeable_visitor_count in v_booking_overview.
- A full Maven package could not be executed in the sandbox because the Maven wrapper attempted to download Maven from the internet and network access is unavailable. This is an environment limitation rather than a detected source-code error.

Suggested final acceptance test

- Use a package with known price, create a booking containing at least one under-5 child and confirm the displayed estimate excludes that child from per-person price.
- Confirm booking as Operator, perform card or approved bank payment, assign team, issue permit, check in, check out, then verify Driver completion becomes available only after checkout.
- Toggle dark mode on Sign In and verify every demo login role/email remains readable.
- Open Park Officer dashboard and verify the tab reads Park Compliance.
- Confirm Kavi faces forward, waves hello, shows the Hi bubble, and still opens the assistant panel.

Final deliverables

- Full updated Safari Lanka project ZIP.
- Full standalone MySQL SQL file.
- This consolidated PDF change register and workflow document.

End of document